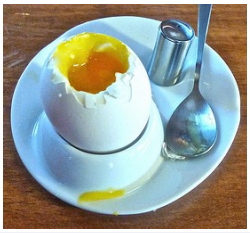
Part of keeping healthy is reaching a balance between the energy you take in as food and the energy you use up.

Task

Use the information given about Sam's diet and activity to answer the questions.

Breakfast

Glass of orange juice 153kJ	Bowl of cornflakes sugar and milk 700kJ	Slice of bread and butter 520kJ	Boiled egg 300kJ
			

1. How much energy would this breakfast give?.....

A table to show how much energy is needed (approximately) for some activities per hour.

Activity	Kilojoules per hour
lying down	320
sitting	360
standing	400
walking quickly	1,280
running	1,640

2. This is Sam's morning. Complete the table, using the information above to show the energy Sam used.

Activity	Time (minutes)	Energy used (kJ)
walked to school	30	
stood around talking	15	
sat in lessons	120	
played sport (running)	15	

3. How much energy did Sam use in total?.....

4. Did the breakfast meet Sam's energy needs?..... Why do you think this?

.....

5. Write down another factor, (other than level of activity), that would affect Sam's energy needs.

.....

6. When lying down, what would the body use the 320KJ of energy for? List 3 things

a..... b..... c.....

Answers

1. 1673 KJ
2.

Activity	Time (minutes)	Energy used (kJ)
Walked to school	30	**640**
Stood around talking	15	**100**
Sat in lessons	120	**720**
Played Football (running)	15	**410**

3. 1870 KJ
4. No because the energy taken in was less than the energy used. (It is only a small snapshot of Sam's diet so difficult to draw a conclusion either way.)
5. Body mass, external temperature, age, gender, proportion of muscle etc
6. Growth, replacing cells, automatic body functions e.g. heart contraction, breathing, digestion, nervous system, making new compounds e.g. enzymes, hormones

Image Credits

© Larry & Ted Page 2010 http://www.flickr.com/photos/96029422@N00/4301970315/